

issues

Issues

Revision: 41 **Last modified:** 2026-06-02T01:13:03Z **Description:** Active workable-item registry for Helix Cluster OS **Authority:** Constitution §11.4.93 (workable-items DB single source of truth) **Generated-by:** scripts/docs/db_to_md.py (DB is canonical; edit via cmd/hxc-registry, not by hand)

Total active items: **424**. Canonical source: data/hxc_registry.db. Full per-item detail (description, closure criteria, required test types, source refs) lives in the DB and docs/research/_ledger/*.json.

Foundation (anti-bluff) (5 active)

HXC	Type	Pri	Status	Title
HXC-016	Task	P1	Queued	
HXC-017	Task	P1	Queued	
HXC-018	Task	P1	Queued	
HXC-019	Task	P1	Queued	
HXC-020	Task	P1	Queued	

MVP (11 active)

HXC	Type	Pri	Status	Title
HXC-1117	Task	P0	Queued	Define and generate proto service stubs (node/session/scheduler/health/auth)
HXC-1100	Feature	P1	Queued	Implement real Bazel RBE build execution (internal/build)

HXC	Type	Pri	Status	Title
HXC-1119	Feature	P1	Queued	Implement Avro event schemas and schema-validated routing (Event Bus)
HXC-1140	Task	P1	Queued	Publish v1.0.0-dev-mvp release across all modules with packaging
HXC-1113	Bug	P2	Queued	Remove dangling pkg/gpuattest placeholder and reuse security/pkg/gpuattest
HXC-1126	Feature	P2	Queued	Implement LLM Brain advisory engine with LLMsVerifier gate (internal/llm)
HXC-1128	Feature	P2	Queued	Implement Health Monitor with eBPF probes and LSTM failure prediction
HXC-1129	Feature	P2	Queued	Implement CRIU/DMTCP-based session live migration orchestration
HXC-1139	Research	P2	Queued	Author TLA+ specifications for consensus and scheduling concurrency
HXC-1143	Feature	P2	Queued	Implement distcc/icecream distributed C/

HXC	Type	Pri	Status	Title
HXC-1145	Docs	P2	Queued	C++ compilation and ccache layer Author MVP architecture and component-specification documentation artifacts

Phase 2 (14 active)

HXC	Type	Pri	Status	Title
HXC-1146	Docs	P0	Queued	Document node-provisioning boundary: project does NOT implement jailbreak/exploit/DRM-circumvention
HXC-905	Feature	P0	Queued	SQLite HXC Registry
HXC-1147	Feature	P1	Queued	Implement internal/console/linux_boot.go BootCoordinator state machine
HXC-1153	Feature	P1	Queued	Implement pkg/resources Linux GPU reader via /sys/class/drm (NVML-free)
HXC-1159	Task	P1	Queued	E2E: a Linux node joins the cluster mesh and becomes schedulable
HXC-906	Docs	P1	Queued	PHASE_2_ROADMAP.md
HXC-1154	Feature	P2	Queued	Integrate universal Vulkan compute backend (vendor-neutral, no device-specific code)
HXC-1155	Feature	P2	Queued	Implement console/node thermal sink reader via /sys/class/thermal
HXC-1158	Task	P2	Queued	Harden device hardware-identification in internal/console/detector.go (generic, fixture-tested)
HXC-1161	Feature	P2	Queued	E2E: aggregate GPU pool runs llama.cpp inference

HXC	Type	Pri	Status	Title
				across heterogeneous nodes
HXC-1163	Task	P2	Queued	Validate cross-NAT WireGuard mesh hole-punching across real hosts
HXC-1164	Docs	P2	Queued	Author phase_02 architecture diagram (operator console + mesh + node integration)
HXC-1165	Task	P2	Queued	Ensure api/v1/node.proto carries trust_level + thermal fields consumed end-to-end
HXC-1166	Research	P3	Queued	CRIU/DMTCP live process migration (research + spike)

Phase 3 (40 active)

HXC	Type	Pri	Status	Title
HXC-1167	Task	P0	Queued	Build ARM64 cross-compilation toolchain for Helix agent
HXC-1168	Feature	P0	Queued	Implement SBC Linux node agent (Orange Pi 5 Max target, Tier T3)
HXC-1171	Task	P0	Queued	Validate Mali-G610 Vulkan/OpenCL compute on SBC
HXC-1172	Feature	P0	Queued	Scaffold Android Agent APK (Kotlin + NDK + Termux)
HXC-1173	Feature	P0	Queued	Implement Android foreground service framework with START_STICKY
HXC-1174	Feature	P0	Queued	Integrate Android BatteryManager monitoring

HXC	Type	Pri	Status	Title
HXC-1175	Feature	P0	Queued	Implement charging-gated PowerGater scheduling guard
HXC-1178	Feature	P0	Queued	Implement MQTT client for edge work dispatch and status
HXC-1184	Feature	P0	Queued	Implement edge node registration for all tiers (T3-T8)
HXC-1185	Feature	P0	Queued	Implement edge heartbeat with battery/thermal/network telemetry
HXC-1186	Feature	P0	Queued	Implement edge protocol gateway (MQTT/QUIC/WebSocket)
HXC-1187	Feature	P0	Queued	Define EdgeWorkUnit and EdgeWorkResult protobuf schemas
HXC-1197	Feature	P0	Queued	Implement edge output verification (LLMsVerifier/redundant/checksum)
HXC-1205	Feature	P0	Queued	Build edge setup wizard for device onboarding
HXC-1207	Task	P0	Queued	Implement edge device chaos test suite
HXC-1208	Task	P0	Queued	Implement battery/thermal stress testing for edge devices
HXC-1169	Feature	P1	Queued	Implement SBCAdapter for SBC-specific hardware monitoring
HXC-1170	Feature	P1	Queued	

HXC	Type	Pri	Status	Title
				Integrate RK3588 NPU via RKNN Toolkit2 C API
HXC-1176	Feature	P1	Queued	Implement Android Vulkan/ NNAPI compute backend
HXC-1177	Feature	P1	Queued	Implement QUIC transport client for Android (Tier T6)
HXC-1179	Feature	P1	Queued	Scaffold iOS Agent app (Swift, Tier T7 EDGE_DONOR)
HXC-1180	Feature	P1	Queued	Implement iOS Metal compute integration
HXC-1182	Feature	P1	Queued	Implement iOS background execution scheduling (BGAppRefresh/ BGProcessing)
HXC-1192	Feature	P1	Queued	Implement unified NPU backend with ONNX model converter
HXC-1193	Feature	P1	Queued	Implement per-tier workload quantization selection
HXC-1194	Feature	P1	Queued	Integrate MLC LLM universal engine for mobile inference
HXC-1200	Feature	P1	Queued	Implement Jetson CUDA-on-ARM GPU backend
HXC-1203	Task	P1	Queued	Implement Armbian TV-box provisioning path (RK3588/ Amlogic)
HXC-1204	Task	P1	Queued	Implement ADB-over-WiFi fleet

HXC	Type	Pri	Status	Title
HXC-1206	Feature	P1	Queued	provisioning for Android cluster Build APK/IPA distribution system
HXC-1209	Task	P1	Queued	Validate ARM SBC per-watt performance >=80% of x86
HXC-1181	Feature	P2	Queued	Implement iOS CoreML / Neural Engine inference engine
HXC-1183	Feature	P2	Queued	Implement HarmonyOS Agent (ArkTS) with Da Vinci NPU inference
HXC-1201	Feature	P2	Queued	Implement low-bandwidth protocol optimizations (CoAP option)
HXC-1216	Feature	P2	Queued	Implement true embedded Raft consensus (pkg/raft)
HXC-1217	Feature	P2	Queued	Implement internal/gpu cgo vendor backends (CUDA/ROCm/oneAPI/MLX)
HXC-1218	Feature	P2	Queued	Implement session live migration (CRIU/DMTCP)
HXC-1219	Feature	P2	Queued	Implement WireGuard UPnP/NAT-PMP NAT traversal
HXC-1221	Docs	P2	Queued	Document Phase 3 edge multi-platform agent architecture diagrams
HXC-1202	Research	P3	Queued	Implement WebRTC P2P data-channel

HXC	Type	Pri	Status	Title
				transport for edge devices

Phase 4 (49 active)

HXC	Type	Pri	Status	Title
HXC-1222	Feature	P0	Queued	Build K3s orchestration substrate with RuntimeClasses for firecracker/kata/runc
HXC-1223	Feature	P0	Queued	Implement Firecracker microVM provisioner for T1-T3 with vsock and virtio-net
HXC-1224	Feature	P0	Queued	Implement Firecracker golden snapshot create/restore pipeline (<=28ms restore)
HXC-1232	Feature	P0	Queued	Implement unified golden snapshot + COW instant-reset across all simulator backends
HXC-1234	Feature	P0	Queued	Build Rust DST engine: single-threaded SimLoop with virtual clock and seeded PRNG
HXC-1235	Feature	P0	Queued	Integrate Rust turmoil for deterministic simulated TCP/UDP networking
HXC-1242	Feature	P0	Queued	Build Elixir/OTP Chaos Controller GenServer with supervision-tree isolation
HXC-1251	Feature	P0	Queued	Build Virtual Testing Controller OTP supervision tree (one_for_all)
HXC-1252	Feature	P0	Queued	Implement SessionManager with 50-session cap, 2h TTL, resource-quota enforcement
HXC-1253	Feature	P0	Queued	Implement DevicePool GenServer with tier-to-simulator dispatch and health checks
HXC-1256	Feature	P0	Queued	Expose REST API for session CRUD, device provisioning and snapshot restore

HXC	Type	Pri	Status	Title
HXC-904	Feature	P0	In progress	Phase 4 Build Service
HXC-1225	Feature	P1	Queued	Implement QEMU/KVM ARM64 virt provisioner for T6 RK3588-approximate SBC
HXC-1226	Feature	P1	Queued	Integrate Cuttlefish/CrosVM provisioner for T5 Android AOSP devices
HXC-1228	Feature	P1	Queued	Implement Docker + binfmt_misc protocol stubs for T7 iOS and T8 HarmonyOS
HXC-1230	Feature	P1	Queued	Implement automated tier detection validating host KVM/ARM64/binfmt capabilities
HXC-1233	Feature	P1	Queued	Implement qcow2 copy-on-write overlay management with 10-deep chain limit
HXC-1245	Feature	P1	Queued	Implement 3 time fault injectors (clock skew, clock freeze, monotonic drift)
HXC-1250	Feature	P1	Queued	Integrate Chaos Mesh CRDs (NetworkChaos, TimeChaos, StressChaos, DNSChaos)
HXC-1257	Feature	P1	Queued	Build Phoenix LiveView real-time test dashboard
HXC-1258	Feature	P1	Queued	Implement MetricsCollector exporting 15+ chaos Prometheus metric series with OpenTelemetry tracing
HXC-1259	Feature	P1	Queued	Implement HelixQA automatic challenge generation from test outcomes
HXC-1262	Feature	P1	Queued	Integrate CI/CD quality gates for GitHub Actions / GitLab CI / Jenkins
HXC-1263	Feature	P1	Queued	Build WebAssembly plugin host on Wasmtime Component Model with WIT bindings
HXC-1264	Task	P1	Queued	Define WIT interfaces for device-simulator, workload-generator, fault-injector, metrics-exporter
HXC-1269	Feature	P1	Queued	

HXC	Type	Pri	Status	Title
				Build cmd/helix-testd daemon as OTP controller node entryptpoint
HXC-1273	Feature	P1	Queued	Implement vsock host-guest control channel for chaos-resilient telemetry
HXC-1284	Task	P1	Queued	Benchmark Firecracker 5,000+ microVMs per host with KSM and memory overcommit
HXC-1285	Task	P1	Queued	Run 72-hour continuous chaos soak proving no resource leaks or degradation
HXC-1286	Docs	P1	Queued	Execute Production Readiness Review with 80-item checklist (>=95% complete)
HXC-1287	Task	P1	Queued	Run full 8-tier matrix (160 nodes) end-to-end within 45-minute CI budget
HXC-907	Docs	P1	Queued	PHASE_4_ROADMAP.md
HXC-1227	Feature	P2	Queued	Implement T4 gaming-console protocol-level constrained x86_64 VM
HXC-1246	Feature	P2	Queued	Implement 6 hardware fault injectors (NMI, mem CE/UE, PCIe AER, CPU bit-flip, thermal throttle)
HXC-1266	Docs	P2	Queued	Provide reference plugins (Rust device-sim, Zig device-sim) compiled to wasm32-wasi
HXC-1270	Feature	P2	Queued	Implement Firecracker MicroVM Kubernetes operator and CRD
HXC-1271	Feature	P2	Queued	Deploy WireGuard encrypted inter-host mesh DaemonSet for multi-host clusters
HXC-1272	Feature	P2	Queued	Implement distributed snapshot pool backend (MinIO) and multi-host scheduling
HXC-1274	Feature	P2	Queued	Implement Mininet WAN topology harness for multi-region network-stack testing
HXC-1275	Research	P2	Queued	Author TLA+/PlusCal formal specs for leader

HXC	Type	Pri	Status	Title
				election and scheduler allocation
HXC-1276	Research	P2	Queued	Author Jepsen black-box linearizability test for task scheduling under partitions
HXC-1279	Feature	P2	Queued	Add eBPF/XDP line-rate programmable network fault injection (cilium/ebpf)
HXC-1283	Docs	P2	Queued	Define Corellium / physical-hardware-in-the-loop path for iOS, PS4, RK3588 GPU/NPU
HXC-1288	Task	P2	Queued	Define polyglot interop boundaries (Rust [?] Go [?] Elixir [?] Wasm)
HXC-1289	Feature	P2	Queued	Integrate libcluster automatic BEAM cluster formation via Kubernetes DNS
HXC-1277	Research	P3	Queued	Integrate Shadow/Phantom for unmodified-binary deterministic network simulation
HXC-1280	Research	P3	Queued	Validate gem5 big.LITTLE (O3+Minor) scheduler fidelity for RK3588
HXC-1281	Research	P3	Queued	Evaluate VirGL/Venus virtual GPU for OpenGL/Vulkan workload functional testing
HXC-1282	Research	P3	Queued	Integrate Tart OCI-native macOS/Linux VMs for iOS/macOS CI build-and-test

Phase 5 (31 active)

HXC	Type	Pri	Status	Title
HXC-1293	Feature	P0	Queued	Extend api/v1/node.proto with tier, trust_level, compute_class, arch fields
HXC-1294	Feature	P0	Queued	Create api/v1/resources.proto with npu_tops, fpga_logic_elements, tflops_gpu

HXC	Type	Pri	Status	Title
HXC-1295	Feature	P0	Queued	Extend api/v1/scheduler.proto with tier constraints and power_budget
HXC-1300	Feature	P0	Queued	Extend pkg/resources to probe NPU TOPS, FPGA logic elements, GPU TFLOPS, QPU
HXC-1314	Feature	P0	Queued	Implement pkg/handheld Steam Deck agent (Vulkan compute, battery, thermal) (DEFERRED hw)
HXC-1305	Feature	P1	Queued	Implement device discovery engine: GPU detection (Vulkan/CUDA/ROCm/Mali strategies)
HXC-1306	Feature	P1	Queued	Implement device discovery engine: NPU detection (Rockchip/NVIDIA DLA/Qualcomm/Apple)
HXC-1309	Feature	P1	Queued	Implement capability manifest generation + control-plane negotiation
HXC-1312	Feature	P1	Queued	Build multi-arch universal agent container (amd64/arm64/riscv64) via buildx + distroless
HXC-1313	Feature	P1	Queued	Implement riscv64 cross-compilation pipeline + RVV extension detection (DEFERRED CI)
HXC-1315	Feature	P1	Queued	Package Steam Deck Flatpak agent with systemd integration (DEFERRED hw)
HXC-1316	Feature	P1	Queued	Implement internal/sbc: Jetson TensorRT backend (>=60

HXC	Type	Pri	Status	Title
				TOPS) (DEFERRED hw)
HXC-1317	Feature	P1	Queued	Implement internal/sbc RK3588 NPU + Turing RK1 cluster support
HXC-1320	Feature	P1	Queued	Implement internal/enterprise EPYC/ Ampere auto-provisioning + Coreboot trust detection (DEFERRED hw)
HXC-1321	Feature	P1	Queued	Implement internal/iot OpenWrt router gateway agent (GL-MT6000, Docker, <5% overhead)
HXC-1322	Feature	P1	Queued	Implement internal/iot NAS storage-node agent (Synology/QNAP Docker, dual-role)
HXC-1327	Feature	P1	Queued	Implement gVisor/ Kata per-tier sandbox hardening for UNTRUSTED/ edge nodes (DEFERRED)
HXC-1328	Feature	P1	Queued	Implement cloud-spot AWS/Azure/ GCP IMDS interruption pollers (adapters)
HXC-1329	Feature	P1	Queued	Implement hybrid cloud-on-prem WireGuard mesh joining spot workers to on-prem cluster
HXC-1332	Task	P1	Queued	Phase 5a integration test: 10-node mixed handheld+SBC cluster
HXC-1307	Feature	P2	Queued	Implement device discovery engine: FPGA detection (Xilinx/Intel/Lattice, hard vs soft core)
HXC-1308	Feature	P2	Queued	

HXC	Type	Pri	Status	Title
HXC-1318	Feature	P2	Queued	Implement micro-benchmarker producing normalized per-compute-class scores
HXC-1319	Feature	P2	Queued	Implement pkg/fpga + internal/fpga hard-SoC/soft-core/DPU backends (DEFERRED hw)
HXC-1324	Feature	P2	Queued	Implement bitstream verification + management for FPGA tiers (T11 required)
HXC-1325	Feature	P2	Queued	Implement internal/exotic Groq LPU inference adapter (<100ms TTFT) (DEFERRED cloud)
HXC-1331	Docs	P2	Queued	Implement internal/exotic Cerebras CS-3 large-model inference adapter (DEFERRED cloud)
HXC-1333	Task	P2	Queued	Author 64-device master taxonomy table as machine-readable device catalog
HXC-1323	Feature	P3	Queued	Phase 5b integration test: 8-node arm64+riscv64+fpga heterogeneous cluster
HXC-1326	Research	P3	Queued	Implement webOS smart-TV JS agent over WebSocket (DEFERRED device)
HXC-1330	Docs	P3	Queued	Implement pkg/quantum Qiskit Runtime plugin for async circuit submission (DEFERRED)
				Define and validate 5 reference cluster build recipes

HXC	Type	Pri	Status	Title
				(\$250/\$500/\$1k/\$2k/\$5k)

Phase 6 (46 active)

HXC	Type	Pri	Status	Title
HXC-1344	Feature	P0	Queued	Implement pkg/ swim/hierarchical two-tier LAN+WAN delegate gossip
HXC-1348	Feature	P0	Queued	Implement pkg/ nat traversal STUN client and NAT-type classification
HXC-1349	Feature	P0	Queued	Implement ICE candidate gathering, prioritization and connectivity checks
HXC-1352	Feature	P0	Queued	Implement internal/cell WireGuard inter- cell mesh manager
HXC-1369	Feature	P0	Queued	Implement cmd/ helix-federation binary and helixctl federation CLI
HXC-1384	Task	P0	Queued	Establish multi- cell integration testbed for CLAUDE-1 usability gate
HXC-1339	Feature	P1	Queued	Implement Vector Clock causality type for Tier-2 operational state
HXC-1342	Feature	P1	Queued	Implement pkg/ federation federated API aggregation with partial failure
HXC-1343	Feature	P1	Queued	Implement Phi- accrual failure

HXC	Type	Pri	Status	Title
				detector in pkg/ swim
HXC-1345	Feature	P1	Queued	Implement gateway-relayed cross-cell suspicion propagation
HXC-1346	Feature	P1	Queued	Implement bootstrap/ rendezvous strategies (static, DNS-SRV, DHT, cloud, mDNS)
HXC-1347	Feature	P1	Queued	Implement mDNS/DNS-SD local cell discovery advertiser and browser
HXC-1350	Feature	P1	Queued	Implement TURN relay allocation and fallback chain
HXC-1351	Feature	P1	Queued	Implement QUIC fallback transport with 0-RTT and connection migration
HXC-1353	Feature	P1	Queued	Implement WireGuard zero- downtime key rotation with overlap window
HXC-1357	Feature	P1	Queued	Implement pkg/ spiffe/federation trust-bundle exchange across cells
HXC-1361	Feature	P1	Queued	Implement double-encryption (WireGuard L3 + mTLS L7) data path
HXC-1362	Feature	P1	Queued	Implement pkg/ cilium Cluster Mesh client and global-service propagation
HXC-1363	Feature	P1	Queued	Implement federated service

HXC	Type	Pri	Status	Title
				discovery (cell-local + global registry)
HXC-1364	Feature	P1	Queued	Implement internal/federation Karmada PropagationPolicy engine
HXC-1365	Feature	P1	Queued	Extend pkg/scheduler with latency-aware spot/preemptible scoring
HXC-1367	Feature	P1	Queued	Implement pkg/gitops ArgoCD ApplicationSet federation client
HXC-1370	Feature	P1	Queued	Extend internal/node cell agent: gossip-tier selection + federation identity attestation
HXC-1371	Task	P1	Queued	Enforce etcd-stays-within-cell static-analysis lint gate
HXC-1373	Feature	P1	Queued	Implement split-brain prevention and partition-classification logic
HXC-1380	Feature	P1	Queued	Implement split-brain detection PromQL alerts with runbook links
HXC-1385	Task	P1	Queued	Validate Phase 6 exit-gate KPIs against the live federation
HXC-1386	Task	P1	Queued	Verify sub-phase dependency gates 6a->6b->6c->6d soak conditions
HXC-1354	Feature	P2	Queued	Implement SSH tunnel bridge fallback for

HXC	Type	Pri	Status	Title
				control-plane bootstrap
HXC-1355	Feature	P2	Queued	Implement cloud VPN bridge with provider-specific endpoint discovery
HXC-1356	Feature	P2	Queued	Integrate libp2p DHT discovery, GossipSub and circuit relay
HXC-1358	Feature	P2	Queued	Implement OPA/ Gatekeeper cross-cluster federated-trust admission
HXC-1359	Feature	P2	Queued	Implement OPA data-residency / data-sovereignty admission policy
HXC-1360	Feature	P2	Queued	Integrate External Secrets Operator + Vault for federation secrets
HXC-1366	Feature	P2	Queued	Implement cloud bursting auto-scaling burst pool with spot termination handling
HXC-1368	Feature	P2	Queued	Implement CRDT-based config sync for cell-local overrides
HXC-1372	Feature	P2	Queued	Implement per-cell etcd Raft WAN-safe tuning profiles
HXC-1374	Feature	P2	Queued	Implement internal/chaos CE-01..CE-12 chaos experiment suite
HXC-1375	Feature	P2	Queued	Implement Turmoil-style deterministic multi-cluster simulation harness

HXC	Type	Pri	Status	Title
HXC-1377	Docs	P2	Queued	Author Phase 6 FMEA catalog with detection and recovery runbooks
HXC-1378	Feature	P2	Queued	Implement hierarchical Prometheus federation across cells
HXC-1379	Feature	P2	Queued	Implement OpenTelemetry cross-cell distributed tracing
HXC-1382	Feature	P2	Queued	Implement Velero tiered disaster recovery backup and restore
HXC-1383	Feature	P2	Queued	Implement five federation topology patterns and block-binding modes
HXC-1376	Docs	P3	Queued	Establish quarterly Game Day federation chaos drill protocol
HXC-1381	Feature	P3	Queued	Build Grafana global-health cell-status-grid dashboard

Phase 7 (39 active)

HXC	Type	Pri	Status	Title
HXC-1387	Feature	P0	Queued	Implement Multi-Raft per-shard consensus manager (G-01, P0)
HXC-1388	Feature	P0	Queued	Implement heartbeat coalescer for Multi-Raft network overhead control
HXC-1390	Feature	P0	Queued	Implement MVCC revision store with B-tree index and time-travel (G-08, P0)
HXC-1392	Feature	P0	Queued	

HXC	Type	Pri	Status	Title
				Implement SLURM-style backfill scheduler over availability timeline (G-02, P0)
HXC-1394	Feature	P0	Queued	Implement CRC16 16,384 hash-slot session router (G-03, P0)
HXC-1395	Feature	P0	Queued	Implement MOVED/ASK redirection and ASKING for in-flight slot migration (G-03)
HXC-1400	Feature	P0	Queued	Implement largest-subcluster-wins voting quorum (G-19, P0)
HXC-1419	Feature	P0	Queued	Build deterministic simulation testing (DST) framework on Turmoil (G-14, P0)
HXC-1420	Feature	P0	Queued	Implement BUGGIFY chaos macros at 25% deterministic fire rate (G-15, P0)
HXC-1389	Feature	P1	Queued	Implement leaseholder fast-path local reads in Multi-Raft
HXC-1391	Feature	P1	Queued	Implement persistent Watch Manager with synced/unsynced groups (G-08 support)
HXC-1393	Feature	P1	Queued	Implement multifactor priority queue for scheduler (age+fairshare+size+QoS)
HXC-1398	Feature	P1	Queued	Add Startup health-probe tier with GPU grace period (G-04, P1)
HXC-1399	Feature	P1	Queued	Implement SWIM majority PFAIL->FAIL two-phase failure confirmation (G-23, P1)
HXC-1401	Feature	P1	Queued	Implement STONITH fencing agents (IPMI/ cloud/shared-disk) (G-19, P1)
HXC-1404	Feature	P1	Queued	Implement N+K failover capacity admission control (G-22, P1)
HXC-1405	Feature	P1	Queued	

HXC	Type	Pri	Status	Title
				Implement gang scheduling all-or-nothing GPU reservation (G-18, P1)
HXC-1407	Feature	P1	Queued	Implement device-plugin / GRES fingerprinting framework (G-11/G-13/G-17, P1)
HXC-1408	Feature	P1	Queued	Implement BOINC-style redundant-execution trust scorer (G-09, P1)
HXC-1410	Feature	P1	Queued	Implement Cassandra 3-layer repair: hinted handoff + read repair + Merkle anti-entropy (G-01 part, P1)
HXC-1421	Feature	P1	Queued	Integrate Porcupine linearizability checking in CI under fault injection (G-16, P1)
HXC-1425	Docs	P1	Queued	Produce Phase 7 phase-by-phase 23-gap matrix as tracked deliverable ledger
HXC-1396	Feature	P2	Queued	Implement Atomic Slot Migration controller for live session migration (G-03, P2)
HXC-1397	Feature	P2	Queued	Add config-epoch conflict resolution for simultaneous failovers
HXC-1402	Feature	P2	Queued	Implement Pacemaker-style four-type constraint engine (G-20, P2)
HXC-1403	Feature	P2	Queued	Implement SCAN stable virtual-IP/DNS client endpoint (G-21, P2)
HXC-1406	Feature	P2	Queued	Implement topology-aware NUMA/NVLink GPU placement scoring (G-10/G-18, P2)
HXC-1409	Feature	P2	Queued	Implement standalone delta-state CRDT package (G-counter/PN-counter/OR-set/LWW-map) (G-01 part, P2)
HXC-1411	Feature	P2	Queued	Implement Informer cache helixcache.Watcher

HXC	Type	Pri	Status	Title
				(list-watch local cache) (G-05, P2)
HXC-1412	Feature	P2	Queued	Implement rate-limited work queue with exponential backoff (G-06, P2)
HXC-1413	Feature	P2	Queued	Implement API Priority & Fairness FlowSchema- >PriorityLevel->Queue (G-07, P2)
HXC-1414	Feature	P2	Queued	Implement NATS leaf- node edge-to-core store- and-forward topology (G-12, P2)
HXC-1415	Feature	P2	Queued	Implement idempotent producer (PID + sequence) for exactly- once messaging
HXC-1416	Feature	P2	Queued	Implement cooperative incremental rebalancing for consumers/ membership
HXC-1418	Feature	P2	Queued	Implement tiered cache hot(memory)/ warm(NVMe)/cold(SSD) data tiers
HXC-1422	Feature	P2	Queued	Build nightly chaos pipeline (pod kill, partition, disk stall, clock skew) (G-15 part)
HXC-1424	Docs	P2	Queued	Produce hardened HelixCluster architecture diagram and component map
HXC-1417	Feature	P3	Queued	Implement embedded KRaft-style Raft quorum to remove external ZooKeeper
HXC-1423	Docs	P3	Queued	Author TLA+ formal specifications for consensus/coordination protocols (P2)

Phase 8 (66 active)

HXC	Type	Pri	Status	Title
HXC-1434	Feature	P0	Queued	

HXC	Type	Pri	Status	Title
				Implement GraVal-style miner attestation handshake controller in pkg/chutes
HXC-1435	Feature	P0	Queued	Implement GraValVerifier.BatchVerify concurrent attestation with pass-rate KPI
HXC-1436	Feature	P0	Queued	Implement GraVal VRAM-threshold (95%) gate in attestation
HXC-1438	Feature	P0	Queued	Implement pkg/chutes MinerController K3s deployment lifecycle
HXC-1439	Feature	P0	Queued	Define ChutesMinerConfig and ValidatorConfig data structures
HXC-1440	Feature	P0	Queued	Deploy PostgreSQL StatefulSet for miner inventory tracking
HXC-1441	Feature	P0	Queued	Deploy Redis pub/sub Deployment for miner event propagation
HXC-1442	Feature	P0	Queued	Deploy GraVal bootstrap privileged DaemonSet for GPU attestation
HXC-1443	Feature	P0	Queued	Deploy miner-api Deployment with NodePort 32000
HXC-1444	Feature	P0	Queued	Deploy Gepetto strategy engine via ConfigMap-backed Deployment
HXC-1445	Feature	P0	Queued	Deploy registry-proxy host-network DaemonSet on port 30500
HXC-1446	Feature	P0	Queued	Implement custom HelixGepetto dual-resource arbitration strategy
HXC-1447	Feature	P0	Queued	Implement dual-workload capacity reservation in internal/gpu
HXC-1450	Feature	P0	Queued	Implement pkg/bittensor wallet (coldkey/hotkey) and TAO balance
HXC-1451	Feature	P0	Queued	Implement Bittensor SN64 subnet registration flow
HXC-1462	Feature	P0	Queued	Author helm/helixcluster-chutes base chart (values.yaml)
HXC-1465	Feature	P0	Queued	Author scripts/chutes-node-prep.sh bare-metal node preparation
HXC-1466	Feature	P0	Queued	Author scripts/chutes-miner-deploy.sh miner deployment + registration
HXC-1491	Task	P0	Queued	Validate end-to-end full request flow integration (workload->client->E2EE->network->miner->response)
HXC-1426	Feature	P1	Queued	Implement pkg/chutes Chutes OpenAI-compatible API Client (non-streaming)
HXC-1427	Feature	P1	Queued	

HXC	Type	Pri	Status	Title
				Implement Chutes API client fallback-model chain on retrieable failure
HXC-1428	Feature	P1	Queued	Implement SSE streaming decoder for Chutes chat completions
HXC-1429	Feature	P1	Queued	Implement Chutes API model-list and user-account/balance queries
HXC-1430	Feature	P1	Queued	Implement model-router default-model resolution by strategy (latency/throughput/quality/cost)
HXC-1431	Feature	P1	Queued	Wire E2EE inference envelope between pkg/chutes and security/pkg/e2ee
HXC-1448	Feature	P1	Queued	Implement GraVal attestation hook in internal/gpu Manager
HXC-1454	Feature	P1	Queued	Implement MarketplaceAdapter interface and Chutes adapter
HXC-1455	Feature	P1	Queued	Implement UnifiedManager.RouteWorkload concurrent pricing + composite scoring
HXC-1463	Feature	P1	Queued	Author chute-deployment.yaml Helm template for inference pods
HXC-1464	Feature	P1	Queued	Author values-models.yaml with 8 pre-configured model deployments
HXC-1467	Feature	P1	Queued	Author scripts/chutes-health-monitor.sh observability + alerts
HXC-1468	Feature	P1	Queued	Author scripts/chutes-verify.sh end-to-end verification
HXC-1469	Feature	P1	Queued	Wire internal/gateway to route internal AI requests through Chutes API client
HXC-1470	Feature	P1	Queued	Replace internal/llm stub Inference with real model router (latency/throughput/quality/cost)
HXC-1471	Feature	P1	Queued	Deploy vLLM inference engine with PagedAttention defaults
HXC-1472	Feature	P1	Queued	Deploy SGLang inference engine with RadixAttention
HXC-1483	Feature	P1	Queued	Extend pkg/metrics + Prometheus/Grafana with TAO/GraVal/throughput dashboards
HXC-1484	Feature	P1	Queued	Extend pkg/health with miner-api and GraVal DaemonSet health checks
HXC-1485	Task	P1	Queued	Author HelixQA Challenge: miner deployment and TAO earnings
HXC-1486	Task	P1	Queued	Author HelixQA Challenge: E2EE inference roundtrip

HXC	Type	Pri	Status	Title
HXC-1487	Task	P1	Queued	Author HelixQA Challenge: multi-marketplace routing decision
HXC-1488	Task	P1	Queued	Author HelixQA Challenge: GraVal attestation pass/fail
HXC-1489	Task	P1	Queued	Validate Chutes API client P99 first-token streaming latency < 500ms
HXC-1432	Feature	P2	Queued	Implement gzip compression pipeline in E2EE inference envelope
HXC-1433	Task	P2	Queued	Reconcile E2EE AEAD spec deviation (AES-256-GCM vs roadmap ChaCha20-Poly1305)
HXC-1437	Research	P2	Queued	Implement GraVal CUDA Proof-of-Consecutive-VRAM-Work kernel (DEFERRED, GPU)
HXC-1449	Feature	P2	Queued	Implement MIG profile management in internal/gpu
HXC-1452	Feature	P2	Queued	Implement Yuma Consensus / metagraph weight queries in pkg/bittensor
HXC-1456	Feature	P2	Queued	Implement RevenueOptimizer.OptimizeAllocation greedy GPU-to-marketplace assignment
HXC-1457	Feature	P2	Queued	Implement io.net marketplace adapter
HXC-1458	Feature	P2	Queued	Implement Akash marketplace adapter
HXC-1459	Feature	P2	Queued	Implement Salad marketplace adapter
HXC-1460	Feature	P2	Queued	Implement pkg/economics RewardDistributor multi-token distribution
HXC-1461	Feature	P2	Queued	Implement RewardDistributor.GetParticipantROI and break-even tracking
HXC-1473	Feature	P2	Queued	Enable AWQ 4-bit quantization as default model format
HXC-1475	Feature	P2	Queued	Deploy Intel TDX + NVIDIA CC TEE stack via sek8s
HXC-1476	Feature	P2	Queued	Implement hybrid PQC TLS (X25519 + ML-KEM-768) node-to-node transport
HXC-1477	Feature	P2	Queued	Implement Cosign/Sigstore image admission controller for K3s
HXC-1480	Feature	P2	Queued	Implement carbon-aware scheduler with per-job energy metering
HXC-1481	Feature	P2	Queued	Implement EU AI Act compliance documentation pipeline
HXC-1482	Feature	P2	Queued	

HXC	Type	Pri	Status	Title
				Implement export-control tier verification (country-code KYC) at node onboarding
HXC-1490	Task	P2	Queued	Achieve >60% line coverage on pkg/chutes, pkg/marketplace, pkg/e2ee (Codecov gate)
HXC-1453	Feature	P3	Queued	Implement child-hotkey delegation support in pkg/bittensor
HXC-1474	Feature	P3	Queued	Deploy TurboDiffusion video-generation engine
HXC-1478	Feature	P3	Queued	Implement LUKS-encrypted root for miner nodes
HXC-1479	Feature	P3	Queued	Implement Cilium egress control and Watchtower integrity challenges

Phase 8B (68 active)

HXC	Type	Pri	Status	Title
HXC-1492	Feature	P0	Queued	Define pkg/pool GPUProvider interface as single integration seam
HXC-1493	Feature	P0	Queued	Implement GPUTier type with priority ordering (Local..Decentralized)
HXC-1494	Feature	P0	Queued	Implement PoolManager registry with tier-ordered Allocate walk
HXC-1495	Feature	P0	Queued	Implement WorkloadSpec/ GPUDevice/ GPUAllocation/ PoolStatus data structures
HXC-1496	Feature	P0	Queued	Implement candidate filtering (model/ VRAM/cost/label selector)
HXC-1498	Feature	P0	Queued	Implement PriorityScheduler (tier then least-load then cost)
HXC-1502	Feature	P0	Queued	Implement internal/ costbroker

HXC	Type	Pri	Status	Title
HXC-1504	Feature	P0	Queued	ComputeBroker weighted scorer Implement pkg/burst BurstController MONITOR->SPILL->RECOVER hysteresis
HXC-1510	Feature	P0	Queued	Implement pkg/provider/chutes ChutesProvider OpenAI-compatible adapter
HXC-1519	Feature	P0	Queued	Implement pkg/local LocalGPURegistrar with TCO effective-cost
HXC-1521	Task	P0	Queued	Expose security ML-KEM-768 E2EE Session/Transport as pkg/e2ee
HXC-1522	Feature	P0	Queued	Implement pkg/e2ee E2EEProxy outbound remote-GPU traffic proxy
HXC-1530	Feature	P0	Queued	Implement cmd/gpu-pool-manager HTTP+gRPC front-end binary
HXC-1531	Feature	P0	Queued	Implement cmd/burst-controller utilization- driven binary
HXC-1532	Feature	P0	Queued	Implement cmd/e2ee-proxy transparent ML- KEM-768 proxy binary
HXC-1552	Task	P0	Queued	Implement HelixQA BurstChallenge saturate+failover exit gate
HXC-1553	Task	P0	Queued	Implement E2E helix submit -> saturate -> route-to-Chutes demo
HXC-1497	Feature	P1	Queued	Implement global MaxCostPerHour budget cap on allocations
HXC-1499	Feature	P1	Queued	Implement CostAwareScheduler selecting cheapest GPUs meeting SLA
HXC-1500	Feature	P1	Queued	

HXC	Type	Pri	Status	Title
				Implement LatencyAwareScheduler for inference routing
HXC-1501	Feature	P1	Queued	Implement HealthMonitor with 30s checks and auto-failover trigger
HXC-1503	Feature	P1	Queued	Implement ComputeBroker 60s re-scoring loop with price-spike guard
HXC-1505	Feature	P1	Queued	Implement BurstController utilization RingBuffer moving average
HXC-1506	Feature	P1	Queued	Implement BurstController activateBurst capacity estimation + allocation
HXC-1507	Feature	P1	Queued	Implement pkg/burst CostRouter per-workload-type provider scoring
HXC-1508	Feature	P1	Queued	Implement 5-tier fallback chain Chutes->io.net->RunPod->AWS
HXC-1511	Feature	P1	Queued	Implement Chutes retryWithFallback across fallback models on 429
HXC-1514	Feature	P1	Queued	Implement pkg/provider/ionet IONetProvider Ray cluster adapter
HXC-1515	Feature	P1	Queued	Implement pkg/provider/runpod RunPodProvider serverless + warm pool
HXC-1520	Feature	P1	Queued	Implement LocalGPURegistrar nvidia-smi discovery + GPU classification
HXC-1523	Feature	P1	Queued	Implement pkg/e2ee GraValVerifier provider admission gate
HXC-1526	Feature	P1	Queued	Implement pkg/proxy CUDA API interceptor + memory staging

HXC	Type	Pri	Status	Title
HXC-1527	Feature	P1	Queued	Define gpu_proxy.proto gRPC GPUProvider kernel-dispatch service
HXC-1528	Feature	P1	Queued	Implement virtual /dev/ nvidia* device creation proxying to remote GPU
HXC-1529	Feature	P1	Queued	Implement workload- suitability classifier (HPC local vs inference remote)
HXC-1533	Task	P1	Queued	Add ProviderAdapter registration hooks to pkg/gpu
HXC-1534	Feature	P1	Queued	Add GPUTier-aware filter predicate to pkg/ scheduler
HXC-1535	Feature	P1	Queued	Add GPU-tier utilization/cost/ provider-health metrics to pkg/metrics
HXC-1536	Feature	P1	Queued	Register virtual GPU devices into internal/ node cluster resource model
HXC-1537	Task	P1	Queued	Integrate GraValVerifier into pkg/security provider admission
HXC-1540	Feature	P1	Queued	Implement local vLLM + SGLang serving stack deployment
HXC-1541	Feature	P1	Queued	Implement GPU-proxy DaemonSet advertising virtual nvidia.com/gpu
HXC-1548	Feature	P1	Queued	Implement CostTracker monthly cost report vs AWS on-demand
HXC-1549	Task	P1	Queued	Author Helm chart for Phase 8B GPU pool + burst deployment
HXC-1551	Task	P1	Queued	Build Prometheus/ Grafana Phase 8B tier- utilization + cost dashboards
HXC-1554	Task	P1	Queued	Stand up real-provider CI integration (Chutes/ io.net/RunPod)
HXC-1555	Task	P1	Queued	

HXC	Type	Pri	Status	Title
				Capture E2EE usability evidence: tee=true encrypted-traffic gate
HXC-1556	Task	P1	Queued	Benchmark ML-KEM-768 E2EE handshake latency (<1ms target)
HXC-1557	Task	P1	Queued	Run inference p99 <500ms load test via Chutes (1000 requests)
HXC-1509	Feature	P2	Queued	Implement QoS tiers (real-time, interactive, batch, best-effort)
HXC-1512	Feature	P2	Queued	Implement Chutes GetBalance USD balance monitor
HXC-1513	Feature	P2	Queued	Implement Chutes streaming (SSE) chat-completion variant
HXC-1516	Feature	P2	Queued	Implement pkg/provider/aws AWSProvider EC2 Spot adapter
HXC-1517	Feature	P2	Queued	Implement AWS Spot 2-minute interrupt -> CRIU/S3 checkpoint failover
HXC-1524	Feature	P2	Queued	Implement pkg/e2ee TDX AttestationVerifier for sensitive workloads
HXC-1525	Feature	P2	Queued	Implement key rotation CronJob + HSM integration for E2EE keys
HXC-1538	Feature	P2	Queued	Implement @helix.task Go decorator/SDK with lifecycle hooks
HXC-1539	Feature	P2	Queued	Implement intelligent ModelRouter (latency/throughput/cost/tee/balanced)
HXC-1543	Feature	P2	Queued	Implement KEDA queue-depth/custom-metric autoscaling
HXC-1545	Feature	P2	Queued	Implement CRIU GPU-context checkpoint/restore + graceful degradation

HXC	Type	Pri	Status	Title
HXC-1547	Feature	P2	Queued	Implement TCO calculator for hybrid own+burst economic model
HXC-1550	Task	P2	Queued	Author Docker Compose development stack for Phase 8B
HXC-1558	Task	P2	Queued	Run 48-hour load test and 7-day soak (>99.9% success)
HXC-1559	Task	P2	Queued	Run chaos suite for provider failover and DDoS resilience
HXC-1518	Docs	P3	Queued	Document GCP and Azure provider adapter patterns
HXC-1542	Feature	P3	Queued	Implement bittencert blockchain-backed X.509 identity
HXC-1544	Research	P3	Queued	Implement predictive scaling forecast to pre-warm before peaks
HXC-1546	Feature	P3	Queued	Implement chutes-dropzone self-hosted gateway integration

Phase 8C (55 active)

HXC	Type	Pri	Status	Title
HXC-1560	Feature	P0	Queued	Implement pkg/security/e2ee PQ envelope (ML-KEM-768 + HKDF-SHA256)
HXC-1561	Feature	P0	Queued	Add ChaCha20-Poly1305 AEAD cipher variant to pkg/security/e2ee
HXC-1562	Task	P0	Queued	Add Chutes byte-interop test-vector corpus for pkg/security/e2ee
HXC-1568	Feature	P0	Queued	Implement pkg/gpuattest device-info challenge/response + fingerprint

HXC	Type	Pri	Status	Title
HXC-1569	Feature	P0	Queued	Implement seeded matmul proof-of-GPU-work (PoVW) in pkg/gpuattest
HXC-1570	Feature	P0	Queued	Implement O(1) spot-check verification in pkg/gpuattest
HXC-1571	Feature	P0	Queued	Implement device-sealed encrypt/decrypt (SealForDevice/OpenFromDevice) in pkg/gpuattest
HXC-1574	Feature	P0	Queued	Implement pkg/security/attestation TEE quote construct/verify
HXC-1575	Feature	P0	Queued	Implement Intel TDX quote generation with nonce+cert binding (report_data layout)
HXC-1576	Feature	P0	Queued	Implement NVIDIA GPU attestation evidence verification (NRAS/nvTrust)
HXC-1563	Feature	P1	Queued	Implement bidirectional per-request response keypair in pkg/security/e2ee
HXC-1565	Feature	P1	Queued	Implement streaming SSE E2EE decryption (e2e_init + per-chunk) in pkg/security/e2ee
HXC-1566	Feature	P1	Queued	Implement E2EE instance-discovery + single-use nonce contract
HXC-1567	Feature	P1	Queued	Maintain E2EE length-prefixed framed transport over io.ReadWriter
HXC-1577	Feature	P1	Queued	Implement measured-boot / RTMR3 access-config measurement
HXC-1578	Feature	P1	Queued	Implement attestation-gated key

HXC	Type	Pri	Status	Title
				release (LUKS/ secret) handshake
HXC-1580	Feature	P1	Queued	Implement pkg/ modelintegrity hf_cache_verify gate (SHA-256 + size)
HXC-1583	Feature	P1	Queued	Maintain pkg/ scheduler cost-aware GPU placement plugin
HXC-1584	Feature	P1	Queued	Implement value- multiplier preemption in pkg/ scheduler (Gepetto model)
HXC-1585	Feature	P1	Queued	Implement SKIP LOCKED-style optimistic work- claiming in pkg/ scheduler
HXC-1586	Feature	P1	Queued	Implement attestation-gated scheduler admission predicate
HXC-1590	Feature	P1	Queued	Implement NodeSelector placement-constraint schema in pkg/ resources
HXC-1591	Feature	P1	Queued	Extend pkg/resources GPUInfo with device fingerprint + attested/ thermal state
HXC-1592	Feature	P1	Queued	Implement SUPPORTED_GPUS compute-multiplier catalog + LookupMultiplier
HXC-1594	Feature	P1	Queued	Implement LLMOrchestrator vLLM/SGLang backend WRAP (OpenAI HTTP)
HXC-1595	Feature	P1	Queued	Implement streaming-safe failover (retry only before first byte) in LLMOrchestrator
HXC-1604	Feature	P1	Queued	

HXC	Type	Pri	Status	Title
				Maintain pkg/fiber length-prefixed framed miner<->validator transport
HXC-1605	Feature	P1	Queued	Implement fiber ed25519 signed-identity handshake + stake-gated admission
HXC-1613	Docs	P1	Queued	Document Phase 8C exit-gate evidence matrix (CLAUDE-1 usability proofs)
HXC-1564	Feature	P2	Queued	Implement gzip-before-encrypt payload compression in pkg/security/e2ee
HXC-1572	Feature	P2	Queued	Implement filesystem residency challenge in pkg/gpuattest
HXC-1573	Feature	P2	Queued	Implement multi-GPU node enumeration in pkg/gpuattest
HXC-1579	Feature	P2	Queued	Implement pkg/security/admission OPA-style policy + signed-image (cosign) gate
HXC-1581	Feature	P2	Queued	Implement model/revision anti-cheat verification token (V2 HMAC-SHA256)
HXC-1587	Feature	P2	Queued	Implement bounty/auction economic-placement plugin for scheduler
HXC-1588	Feature	P2	Queued	Implement utilization-aware EWMA candidate ranking for scheduler/orchestrator
HXC-1589	Feature	P2	Queued	Implement Kueue-style suspend-then-admit tiered job admission
HXC-1593	Feature	P2	Queued	

HXC	Type	Pri	Status	Title
				Implement VRAM-capacity ($\geq 95\%$) usable-memory gate in resources/gpuattest
HXC-1596	Feature	P2	Queued	Implement per-task ordered fallback chains with dedup/cap + empty-response detection
HXC-1597	Feature	P2	Queued	Implement thermal pre-warm / scale-from-zero state machine (Therm)
HXC-1598	Feature	P2	Queued	Implement passthrough cords + disconnect-aware upstream abort in LLMOrchestrator
HXC-1601	Feature	P2	Queued	Implement pkg/inferenceproxy correlation-ID + backend audit trail
HXC-1602	Feature	P2	Queued	Implement deterministic keyed-hash anonymization in pkg/inferenceproxy
HXC-1603	Feature	P2	Queued	Implement spoof-proof managed-header sanitization at the recording edge
HXC-1606	Feature	P2	Queued	Implement node identity verify-then-pin (TOFU) keypair-rooted TLS
HXC-1607	Feature	P2	Queued	Implement pkg/marketplace registration -> bounty -> metering -> payout loop
HXC-1608	Feature	P2	Queued	Implement pkg/marketplace/audit commit-then-prove reproducible reconciliation
HXC-1614	Docs	P2	Queued	Document Phase 8C architecture integration diagrams (scheduler<-attest,

HXC	Type	Pri	Status	Title
				orchestrator<-e2ee<- TEE, marketplace)
HXC-1582	Feature	P3	Queued	Implement X25519 ephemeral session- key handshake for verification tokens
HXC-1599	Feature	P3	Queued	Implement Claude/ Responses API-shape adapters for LLMOrchestrator front door
HXC-1600	Feature	P3	Queued	Implement error- classified LLM failover taxonomy + sandbox scheduling- hint contract
HXC-1609	Feature	P3	Queued	Implement scale-to- zero metered hot/cold billing state machine
HXC-1610	Feature	P3	Queued	Implement watchtower-style liveness/integrity prober for served instances
HXC-1611	Research	P3	Queued	Implement Karmada- pattern multi-cluster aggregation over SWIM+Raft+etcd (Phase 6 hook)
HXC-1612	Feature	P3	Queued	Implement OpenAPI request-validation admission middleware for control-plane APIs